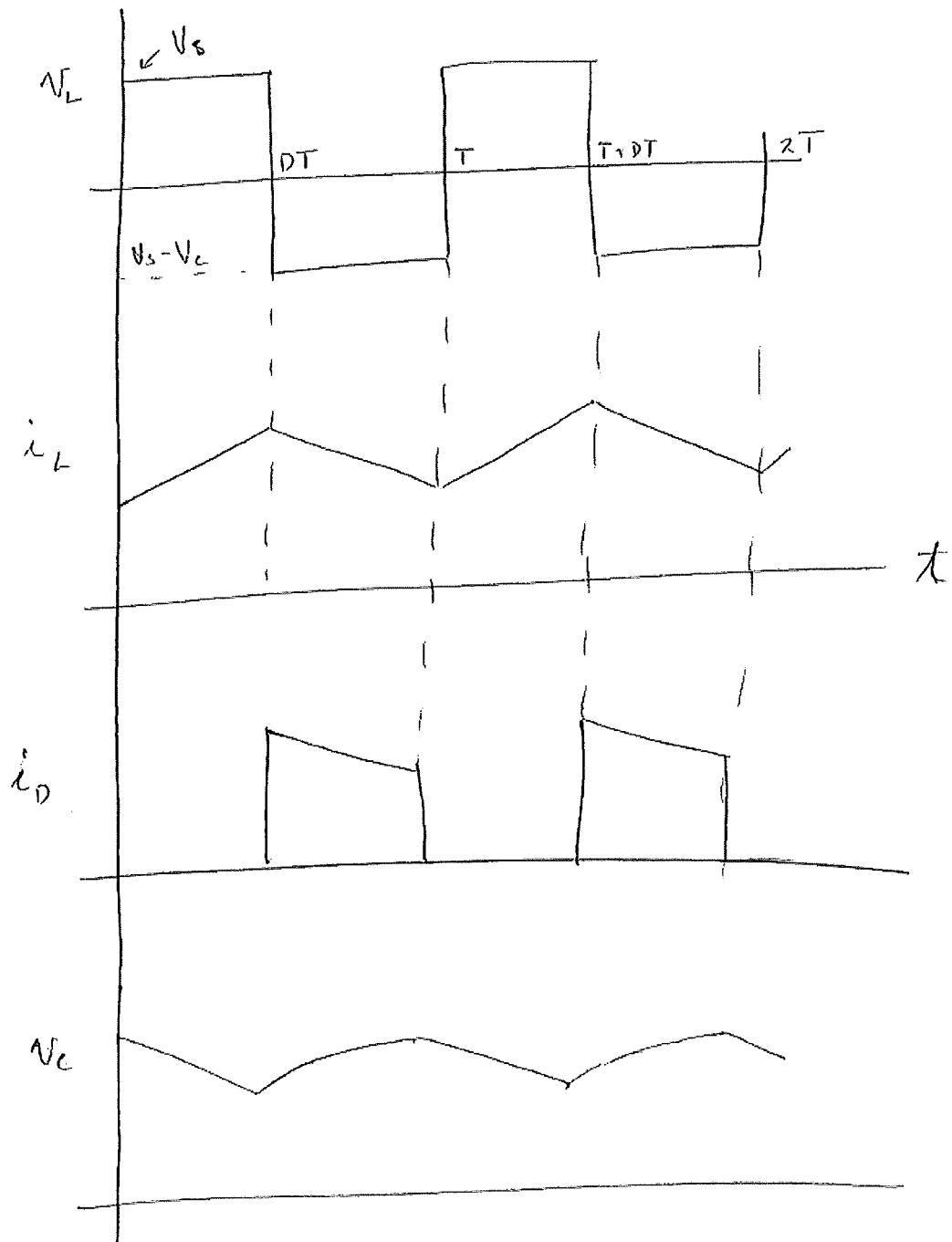


Boost Converter Problems - Solutions

1.

1 a)



1 b) Average inductor voltage should be zero

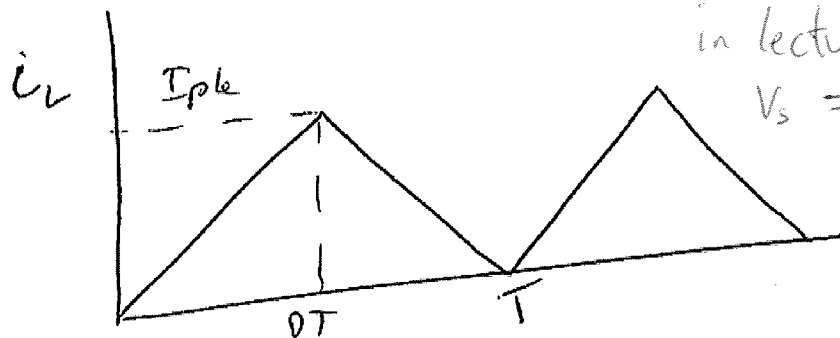
$$\therefore V_S \cdot DT + (V_S - V_C)(1-D)T = 0$$

$$= \cancel{DV_S} + V_S - \cancel{DV_S} - V_C + DV_C = 0$$

$$\therefore V_S - V_C(1-D) = 0$$

$$\therefore \frac{V_C}{V_S} = \frac{1}{1-D} \quad \text{✓}$$

1 c) for just continuous conduction, sketch i_L

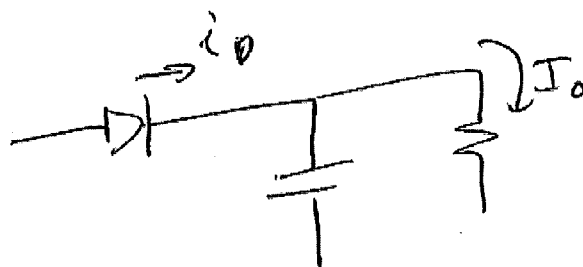


Can solve with method shown in notes, and covered in lecture, noting $V_S = V_C(1-D)$, and

$$\frac{V_O}{I_O} = R.$$

Alternative method

Average current into load side capacitor = 0



i_D is equal to i_L from DT to T

$$\begin{aligned}
 \therefore I_o = I_{oav} &= \frac{1}{T} \int_{DT}^T i_o(x) dt \\
 &= \frac{1}{T} \cdot \frac{I_{pk} \cdot (1-D)T}{2} \\
 &= \frac{I_{pk} \cdot (1-D)}{2} \quad (1)
 \end{aligned}$$

$$\text{and } I_{pk} = \frac{V_s DT}{L} = \frac{V_s D}{f_s L} \quad (2)$$

$$\textcircled{2} \text{ into } \textcircled{1} \quad I_o = \frac{V_s D (1-D)}{2 f_s L}$$

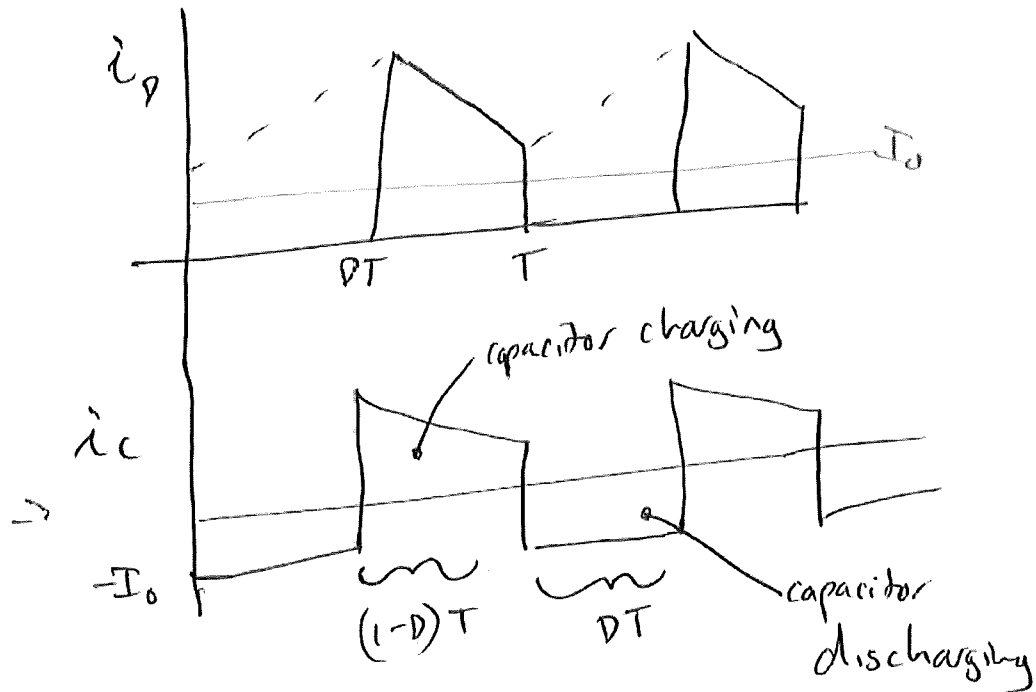
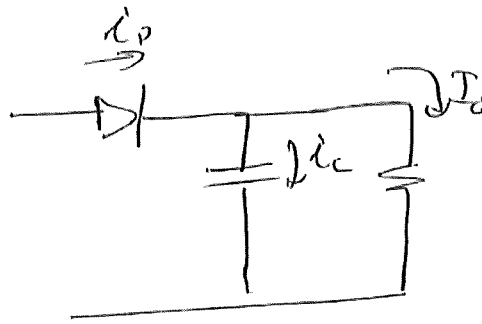
need to convert V_s to V_o $V_o = \frac{V_s}{1-D}$

$$\therefore V_s = V_o (1-D)$$

$$\therefore I_o = \frac{V_o D (1-D)^2}{2 f_s L}$$

$$\therefore R = \frac{V_o}{I_o} = \frac{2 f_s L}{D (1-D)^2} \quad \text{QED}$$

1d)



Using capacitor discharging period $i = C \frac{dv}{dt} = C \frac{\Delta V}{\Delta t}$

$$\therefore \Delta V = \frac{i \Delta t}{C} = \frac{I_o \Delta t}{C} = \frac{I_o D}{f_s C}$$

Rearranging $I_o = \frac{V_o}{R} = \frac{V_s}{(1-D)R}$

$$\therefore \Delta V = \frac{V_s D}{(1-D) f_s R C} = \text{Capacitor voltage ripple}$$